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**This version includes the correction published on 3 March 2020 (doi 10.9999/fake.2019.011c), which adds the trial registration number to section 2.1 and the source of the study iron to the funding statement. The results are unchanged.**

# Dose-response of iron supplementation on fatigue (corrected version)

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## Abstract

Background: Fatigue is the most frequent complaint of adults with iron deficiency without anaemia, yet the amount of oral iron needed to relieve it is not settled. Most trials compare a single dose against placebo, which says little about how much iron is enough. Methods: We conducted a 12-week, double-blind, randomised trial in 150 iron-deficient, non-anaemic adults aged 18 to 55 years, allocated in equal numbers to 20, 40 or 60 mg of elemental iron daily. The primary outcome was the change in a validated fatigue score from baseline to week 12, modelled as a linear function of the daily dose. Results: In 150 iron-deficient adults, fatigue scores fell with increasing iron dose. Ferritin rose in every arm and most in the highest-dose arm, while gastrointestinal complaints were more frequent at 60 mg. Conclusions: Within the range studied, larger daily doses of oral iron produced larger reductions in fatigue, at the cost of more gastrointestinal complaints.

## 1. Introduction

Iron deficiency is the most common nutritional deficiency worldwide. In its early stage, iron stores are depleted while haemoglobin remains within the normal range, and fatigue is the symptom that brings most of these patients to a clinician. Several randomised trials have found that iron given to non-anaemic women with low ferritin reduces self-reported fatigue compared with placebo, but they used different doses, so how the benefit depends on dose is not known. Clinical guidelines recommend daily doses that vary several-fold, and higher doses cause more nausea, constipation and abdominal pain. We therefore randomised iron-deficient, non-anaemic adults to three daily doses of the same oral iron preparation and estimated the change in fatigue per unit of dose.

## 2. Methods

### 2.1 Study design and setting

The trial was a single-centre, parallel-group, double-blind, randomised dose-comparison trial carried out at the outpatient nutrition clinic of Fake University Hospital. The protocol was approved by the institutional ethics committee and registered before the first participant was enrolled (registration FAKE-CTR-2016-0412). All participants gave written informed consent. The trial followed the principles of the Declaration of Helsinki and is reported according to the CONSORT statement for randomised trials.

### 2.2 Participants

Adults aged 18 to 55 years who reported fatigue for at least four weeks were screened. Eligible participants had a serum ferritin below 30 µg/L, a haemoglobin within the sex-specific reference range and a baseline fatigue score of at least 10. We excluded people with a known cause of fatigue other than iron deficiency, a C-reactive protein above 10 mg/L, pregnancy or breastfeeding, and anyone who had taken iron in the previous three months.

### 2.3 Randomisation, blinding and intervention

Participants were randomised in a 1:1:1 ratio to 20, 40 or 60 mg of elemental iron daily, using computer-generated permuted blocks of six stratified by sex. Each participant took three identical capsules each morning, containing iron sulphate or an inert filler in the proportions needed for the allocated dose, for 12 weeks. The iron sulphate was donated by Fake Pharma Ltd; the capsules were prepared by the hospital pharmacy.

**Table 1. Allocated daily doses and capsule composition**

| Arm | Elemental iron per day | Iron capsules | Filler capsules | Participants randomised |
|---|---|---|---|---|
| Low dose | 20 mg | 1 | 2 | 50 |
| Middle dose | 40 mg | 2 | 1 | 50 |
| High dose | 60 mg | 3 | 0 | 50 |

### 2.4 Outcomes and analysis

The primary outcome was the change in the Fatigue Severity Index, a 30-point self-report scale on which higher scores mean more fatigue, from baseline to week 12. It was analysed with a linear mixed model with daily dose as a continuous fixed effect, and the dose-response slope was expressed as the reduction in fatigue score per 10 mg of daily iron, with a 95% confidence interval.

## 3. Results

Of 412 adults screened, 150 were eligible and were randomised, 50 to each arm, and 144 completed the 12-week visit. Fatigue scores fell in all three arms, and the reduction at week 12 was larger in the higher-dose arms. Table 3 gives the dose-response slope: 0.45 points per 10 mg (95% CI 0.30 to 0.60). Adding a quadratic term for dose did not improve the fit of the model.

**Table 3. Dose-response of the Fatigue Severity Index**

| Estimate | Value | 95% CI |
|---|---|---|
| Dose-response slope, points per 10 mg of daily iron | 0.45 | 0.30 to 0.60 |

## 4. Discussion

In this randomised trial in non-anaemic adults with iron deficiency, the reduction in fatigue over 12 weeks grew with the daily dose of oral iron across the range of 20 to 60 mg, with no sign that the benefit levelled off. At the same time, gastrointestinal complaints became more frequent and adherence slightly lower as the dose increased.

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